



From Chat to RAG to MCP

Enhancing Java Applications with AI

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Can we just add ChatGPT to this?

- Pressure to “*add AI*” is everywhere
- Expectations keep rising
- Engineers are caught in the middle

The shift that changed everything

November 2022
ChatGPT goes public

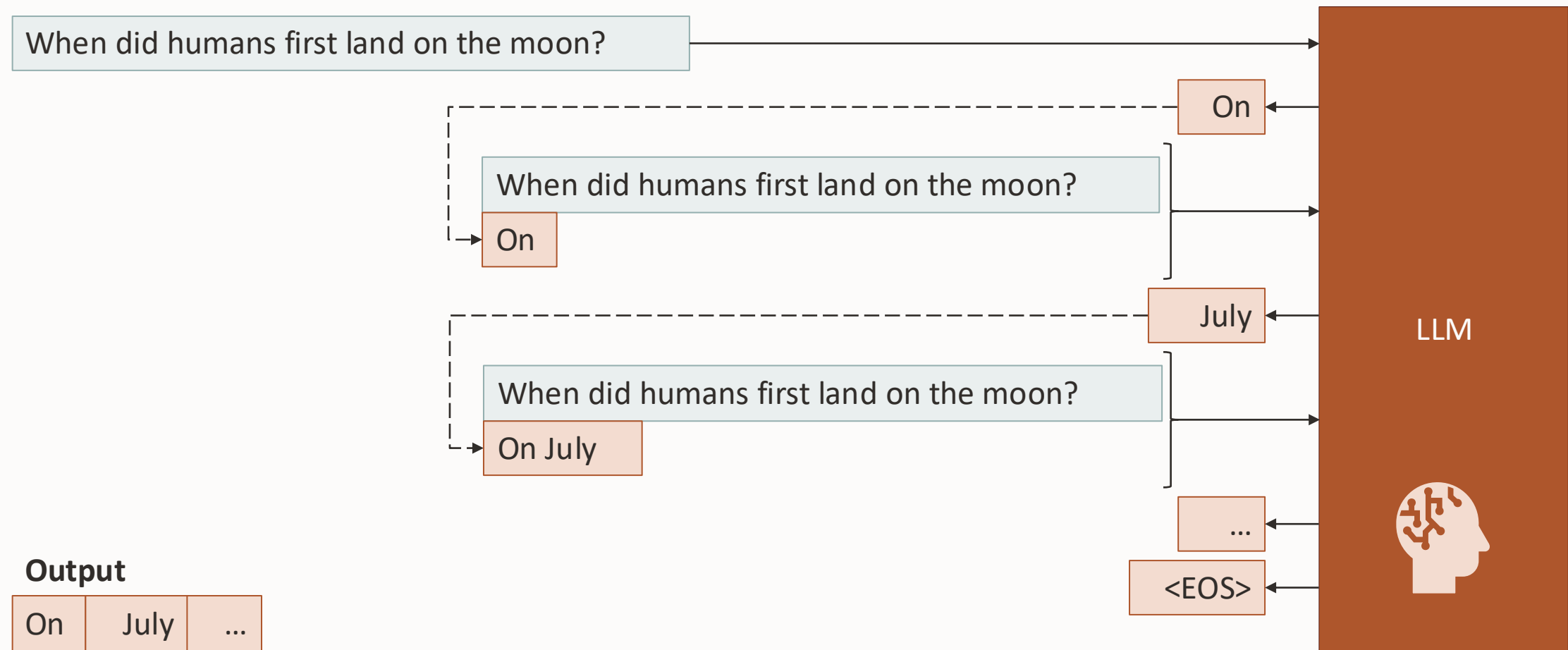


Why Java developers need to understand LLMs

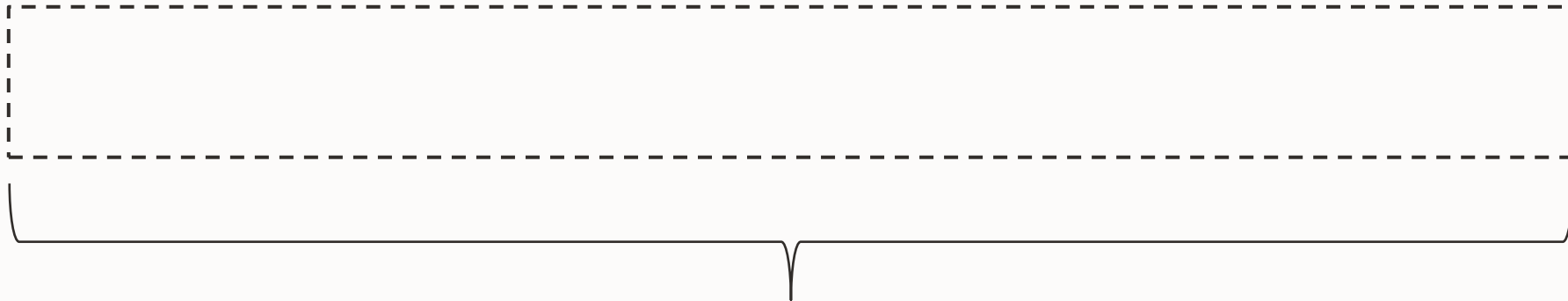
- This is not just another API
- Demos != Production
- It's very difficult to design safeguards for problems you don't yet understand



A practical mental model of how LLMs generate text

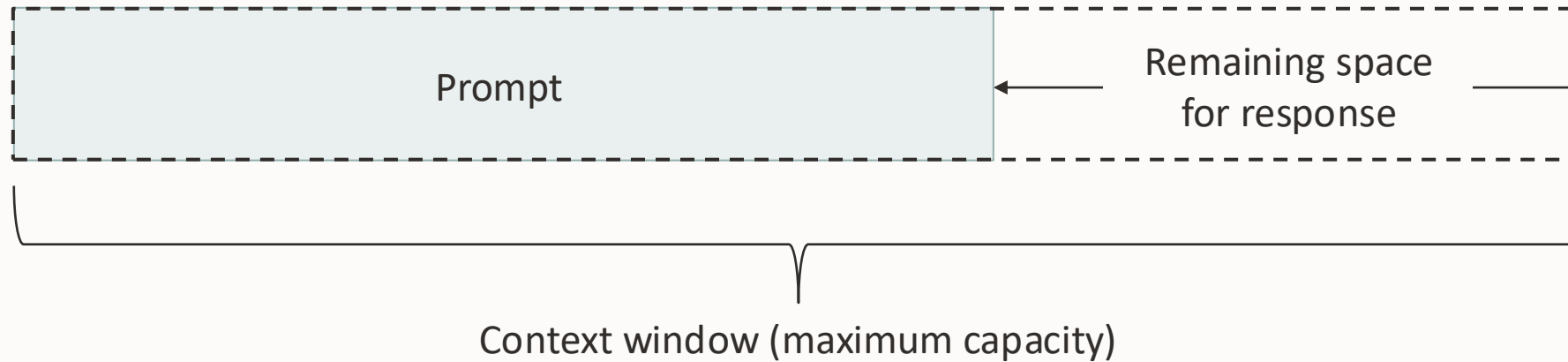


Context window: the model's attention span

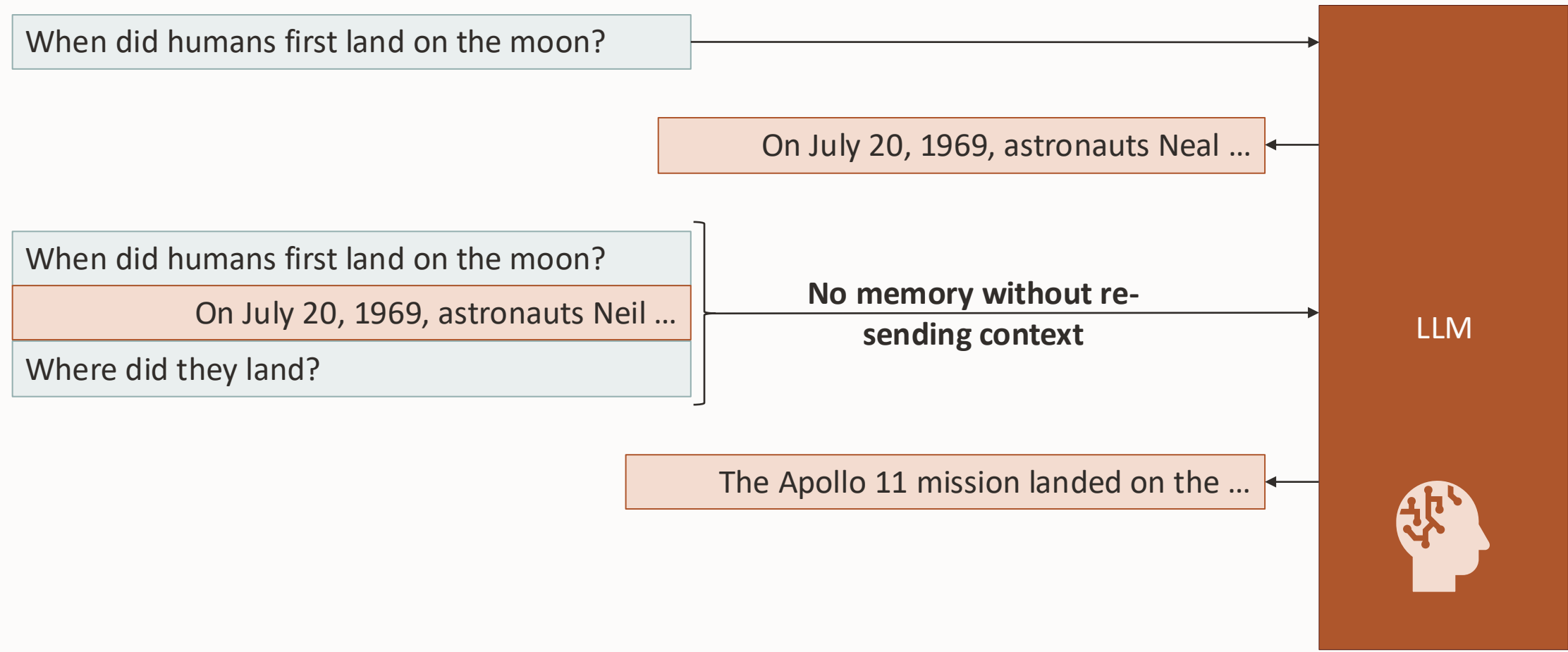


Context window (maximum capacity)

How the context window is used



LLMs always forget



Two practical limitations

Token usage for the same meaning

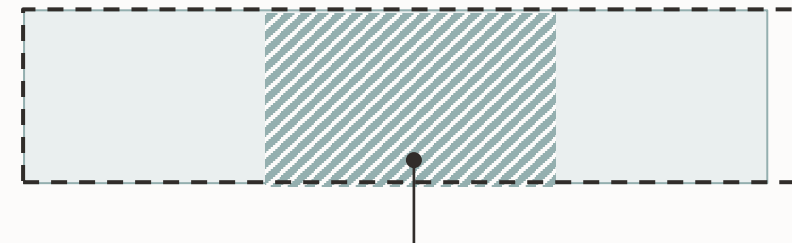
English



Japanese

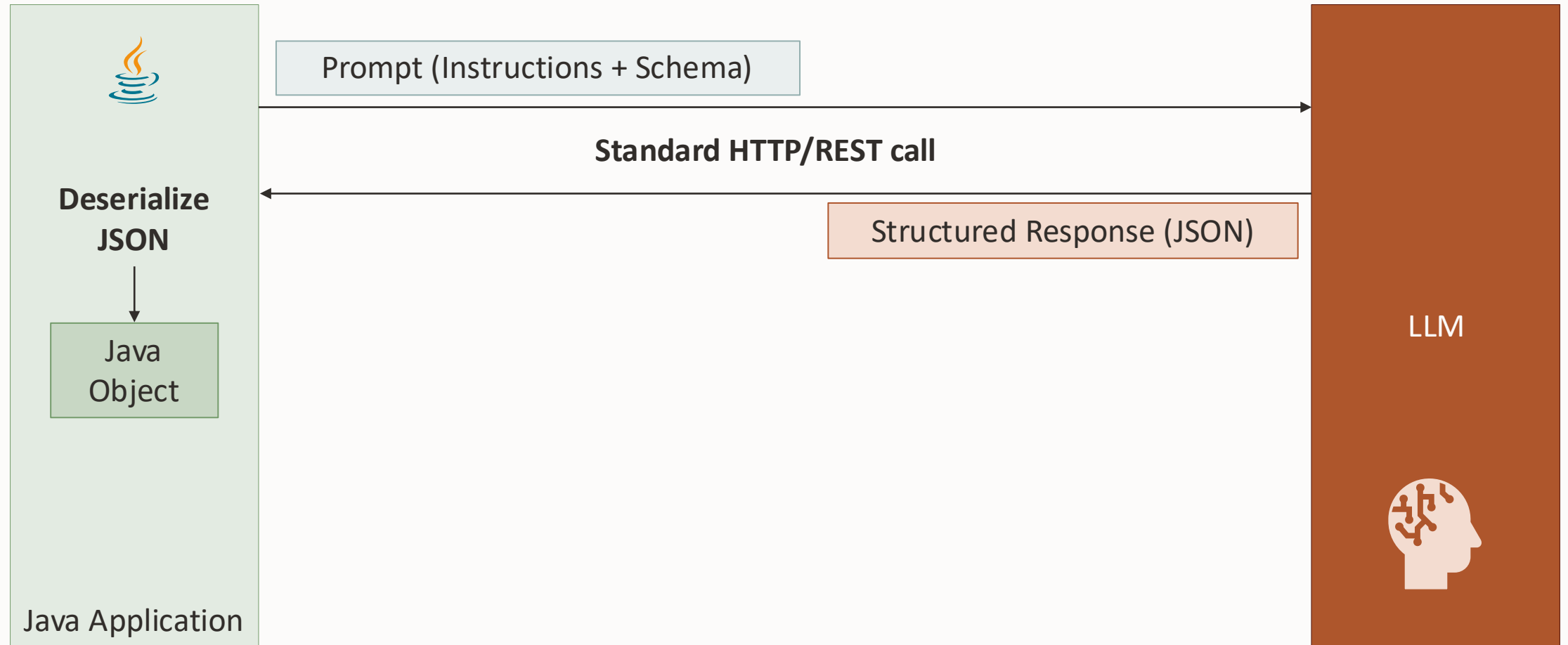


Lost in the Middle



Lower attention / weaker recall

Where does this fit in a Java application?



Example: DIY Store

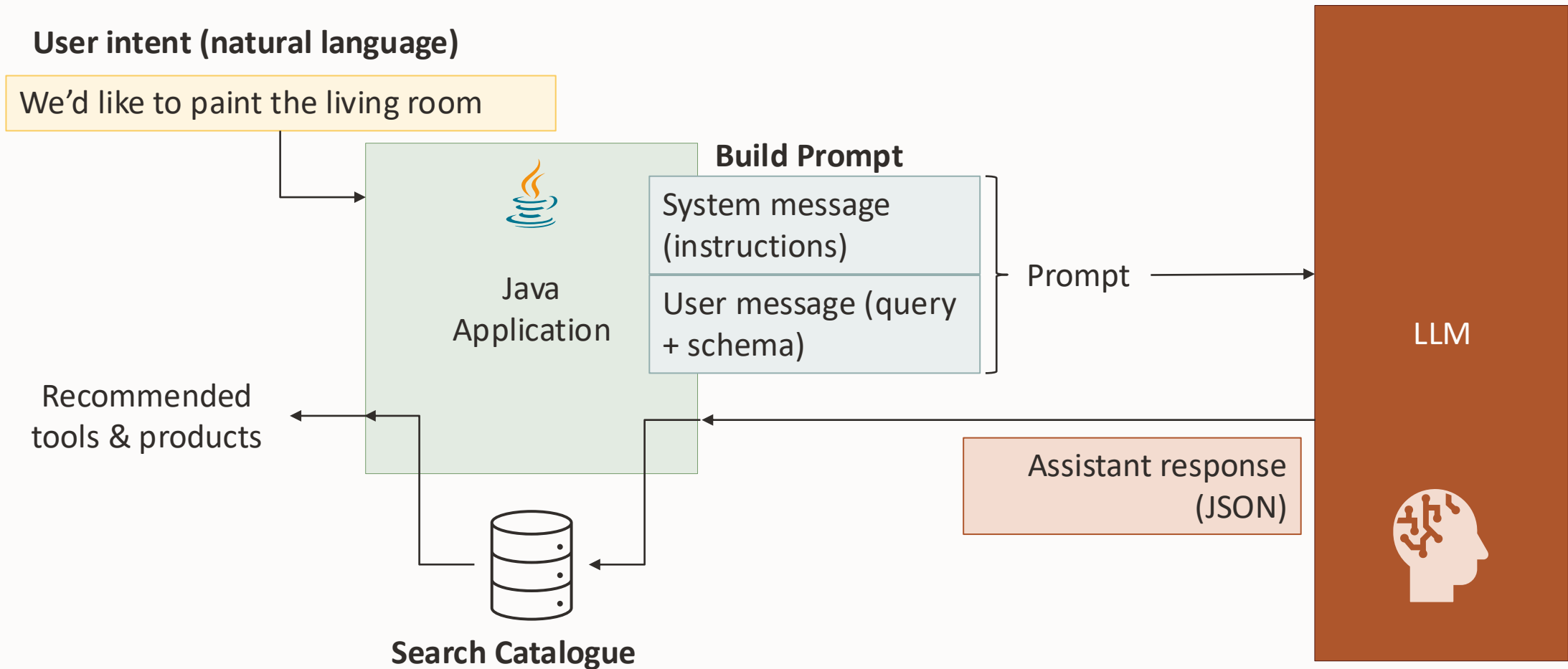


Expresses intent in natural language

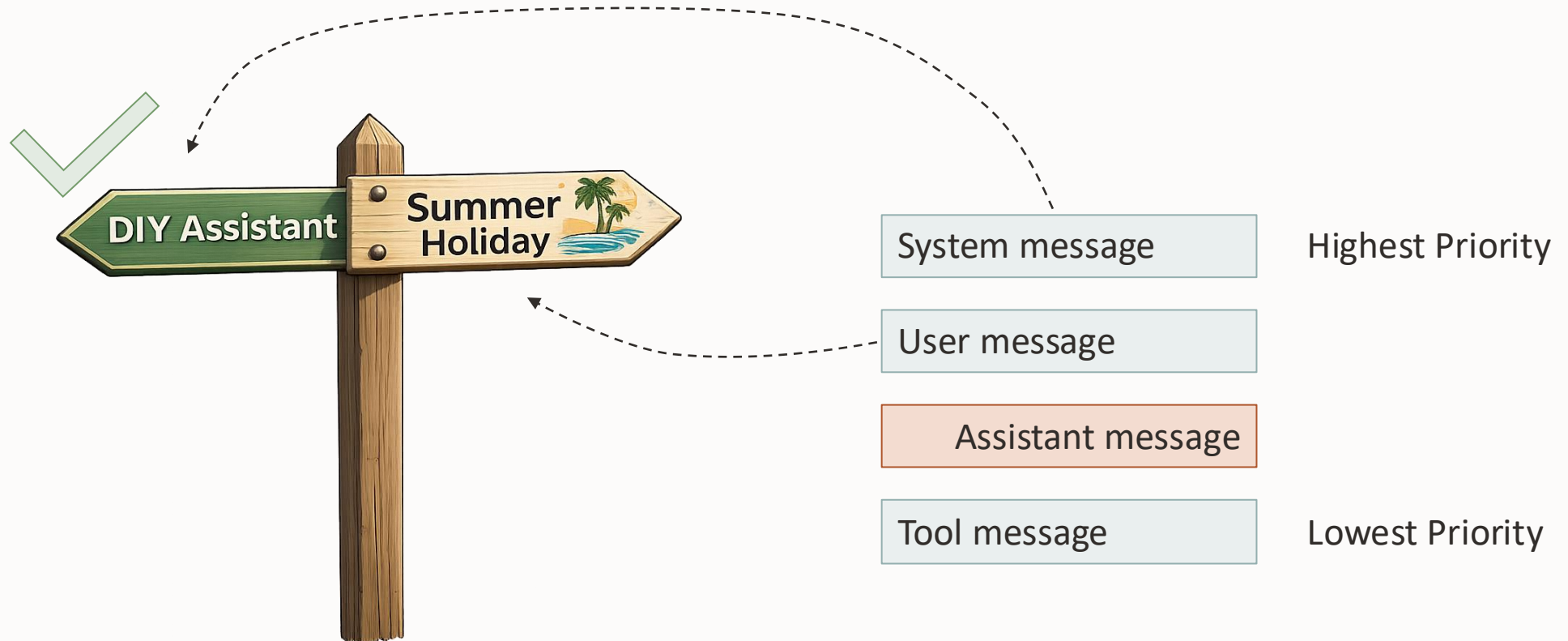


Requires structured product queries

From intent to structure



Why system messages matter



Calling the model from Java

We'd like to paint the living room

```
public Items listRecommendations(final String userMessage) {  
    return client.prompt()  
        .system(spec -> spec.text(systemMessage))  
        .user(spec -> spec.text(userMessage))  
        .call()  
        .entity(Items.class);  
}
```

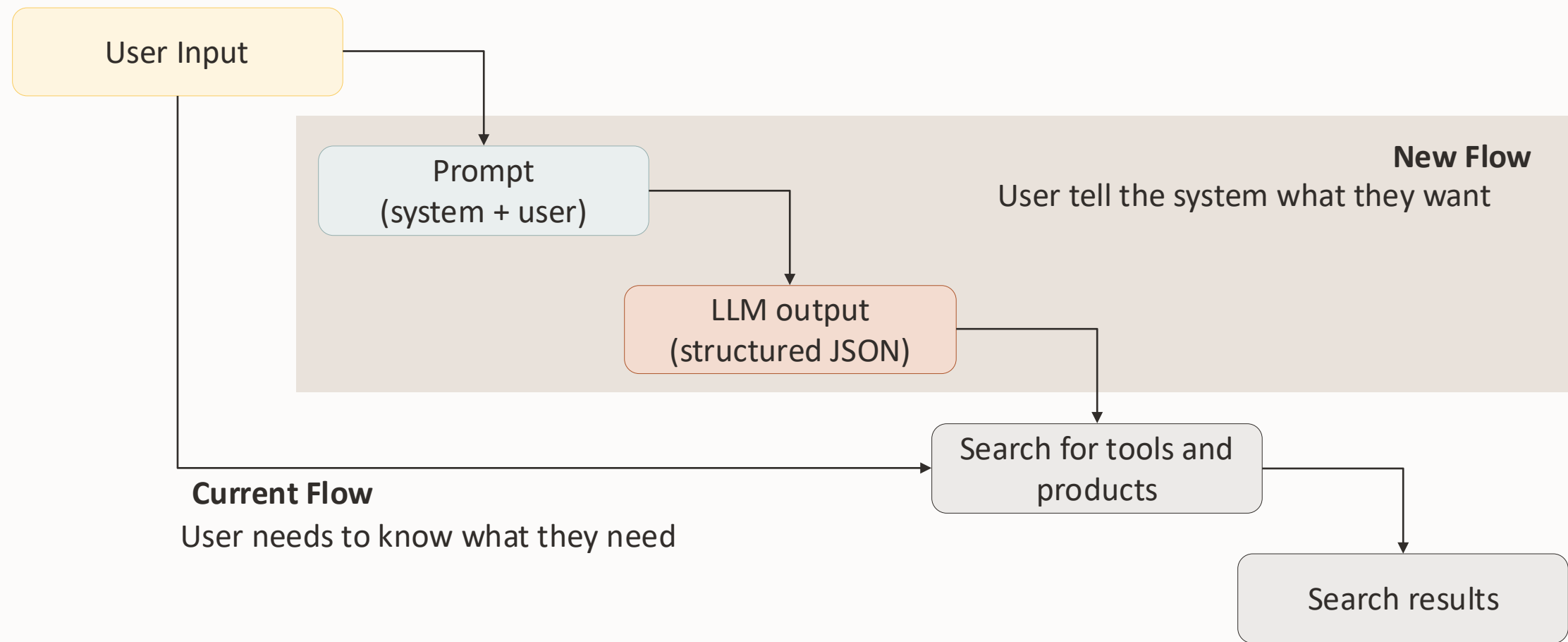
Represents the shape of the output

```
public record Items(List<String> items) {}
```

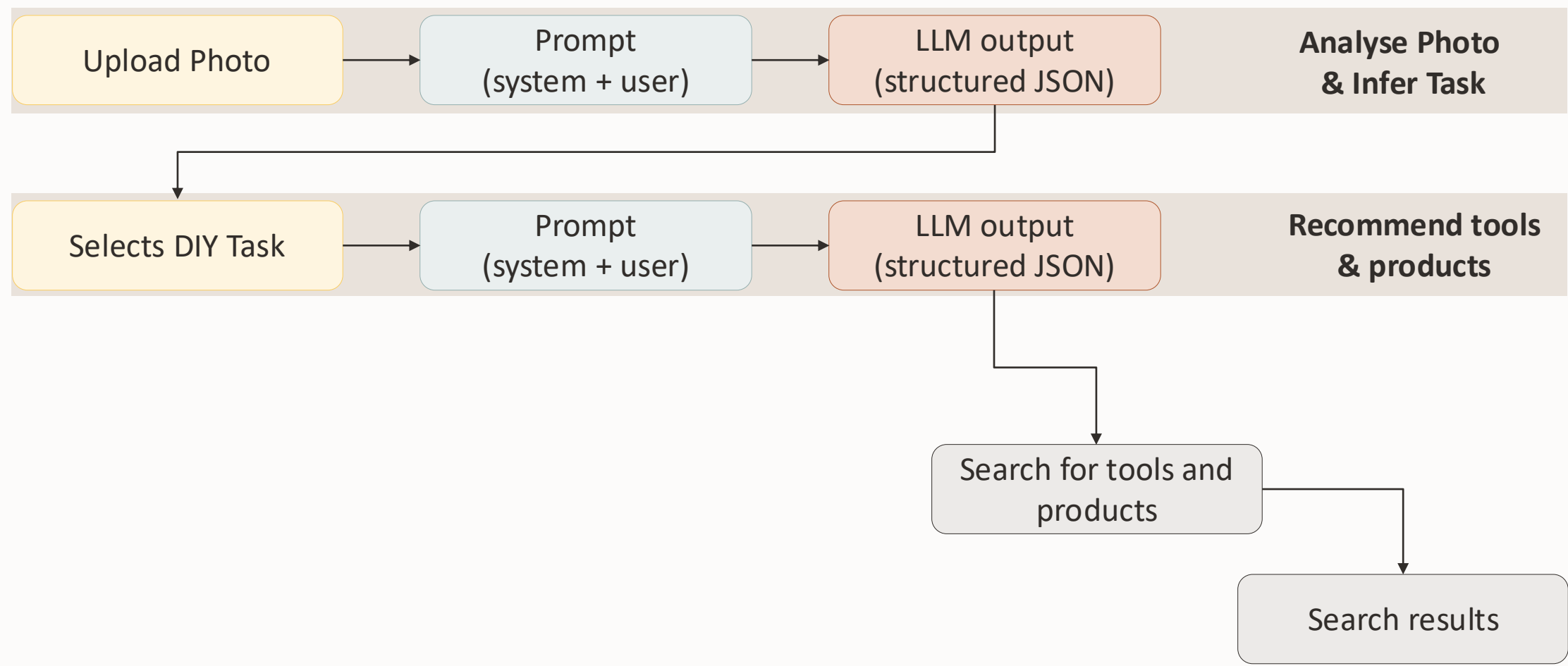
User intent

Java Object

Closing the loop



From text to multimodal interaction

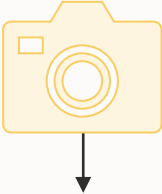


Adding images to user messages

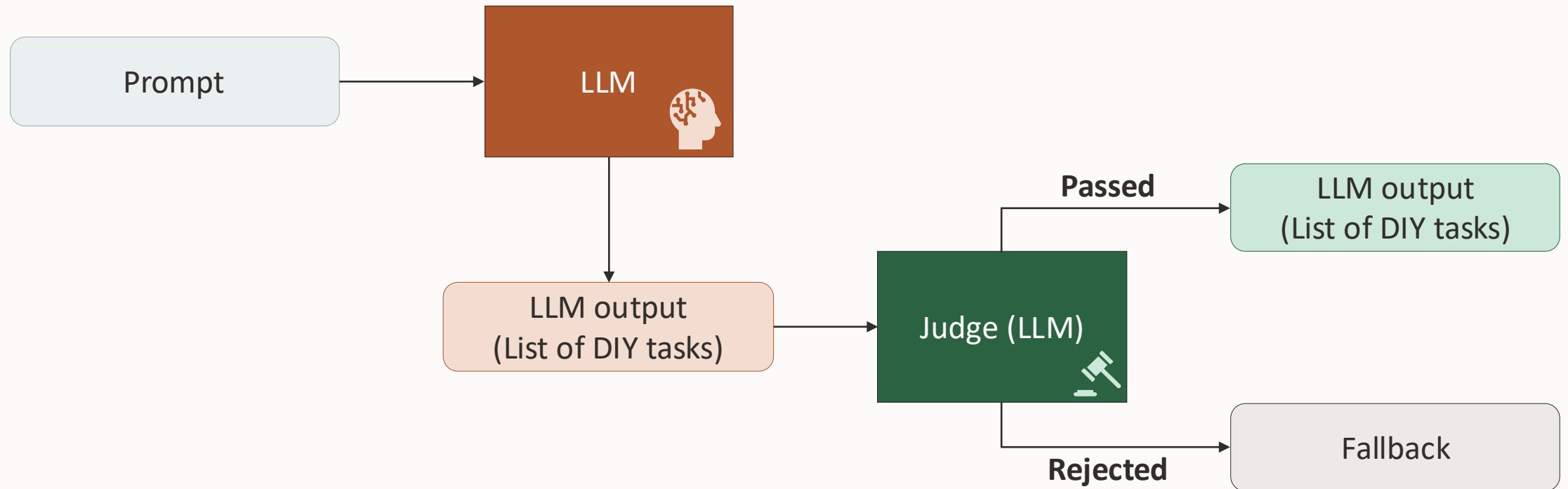
```
final Media media = Media.builder()
    .data(new FileSystemResource(photo))
    .mimeType(MediaTypeFactory.getMediaType(new FileSystemResource(photo))
        .orElse(MediaType.APPLICATION_OCTET_STREAM))
    .build();
```

The image is attached to the user message

```
return client.prompt()
    .system(s -> s.text(systemMessage))
    .user(u -> u.text(taskMessage).media(media))
    .call()
    .entity(PossibleTasks.class);
```



LLM as a Judge



Expanding the interaction model

```
final OpenAiAudioTranscriptionOptions options  
    = OpenAiAudioTranscriptionOptions.builder().build();  
final AudioTranscriptionPrompt prompt  
    = new AudioTranscriptionPrompt(audioResource, options);
```



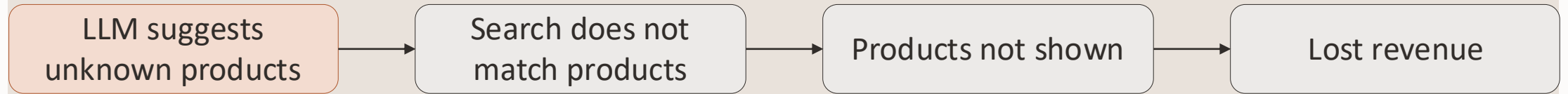
```
final String transcription = transcription.call(prompt)  
    .getResult()  
    .getOutput();
```

Sent as part of the prompt to the model

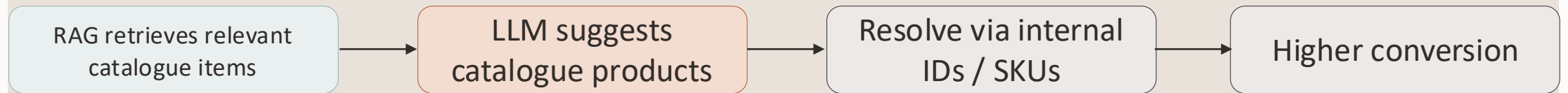


Why grounding the model matters

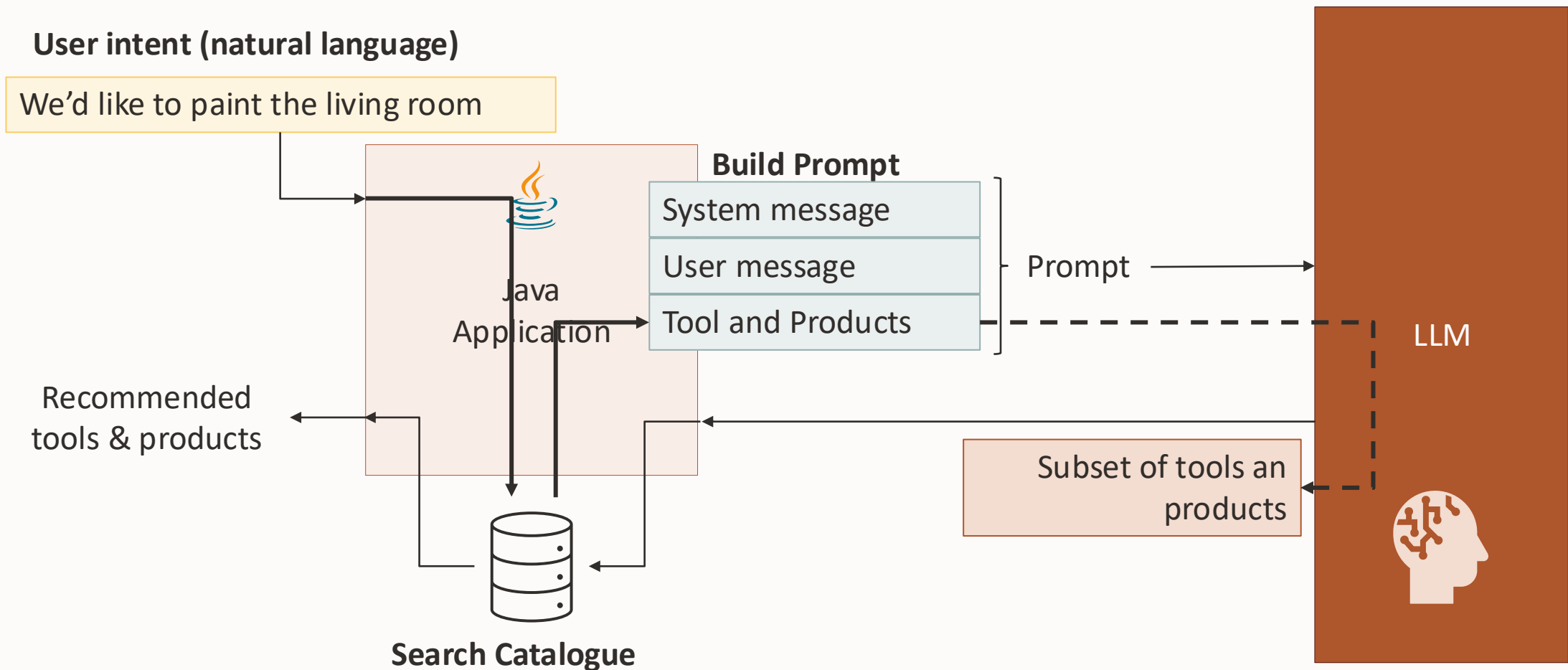
Ungrounded results



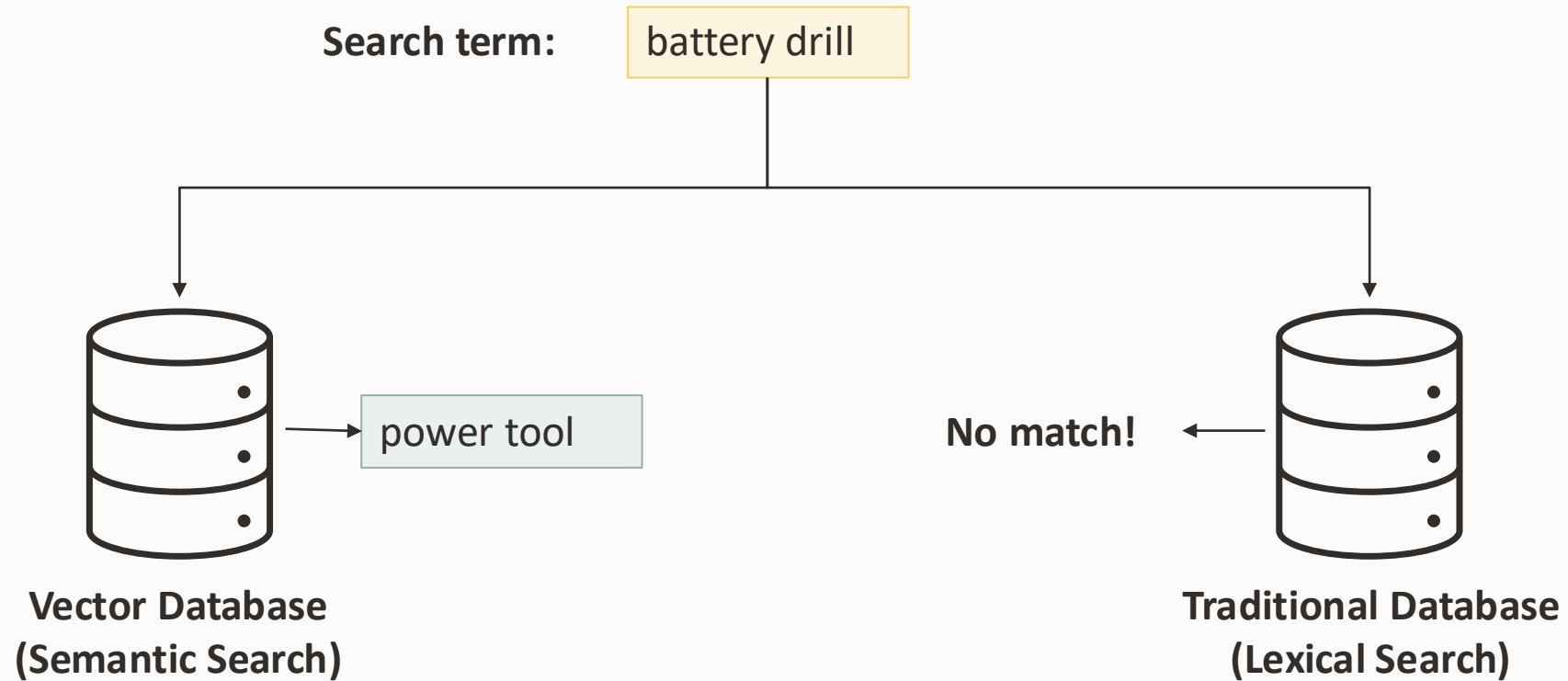
Grounded results



From Prompting to Retrieval: The RAG Pipeline



How RAG finds relevant knowledge



Embedding, embedding model, and vector

Input

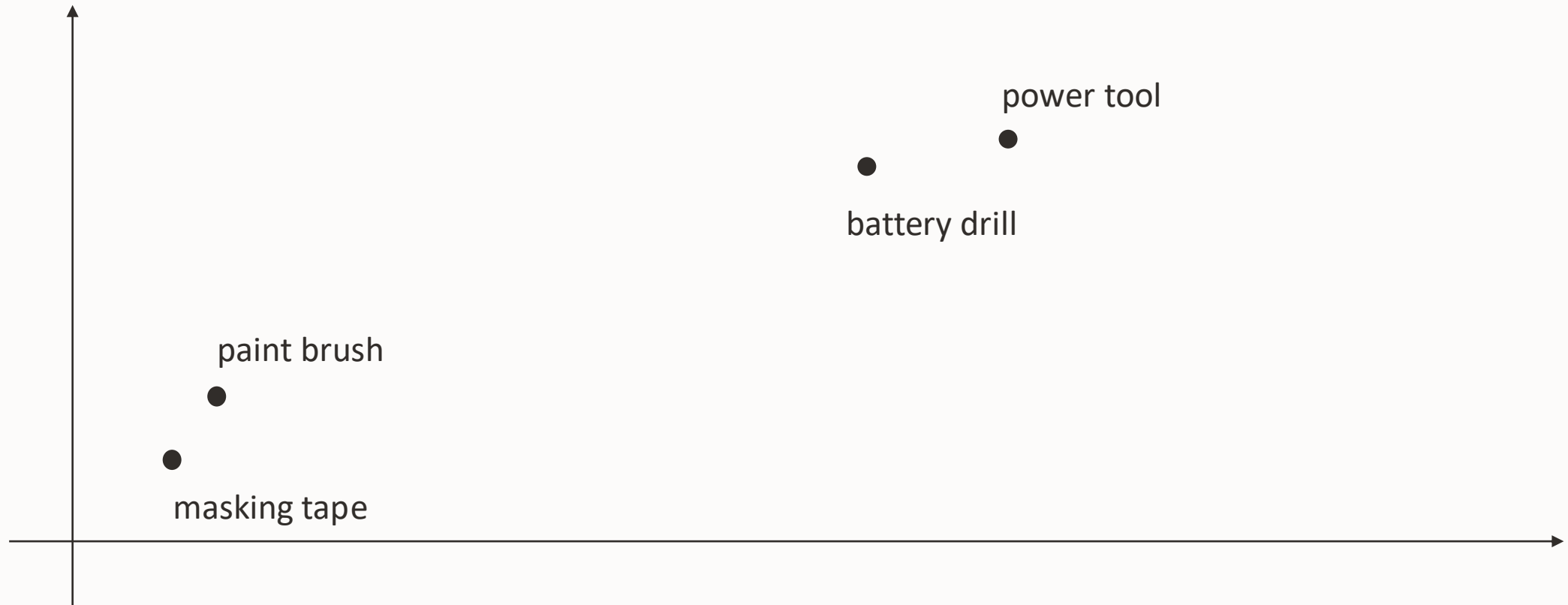
A versatile power drill designed for drilling holes and ...

Embedding Model

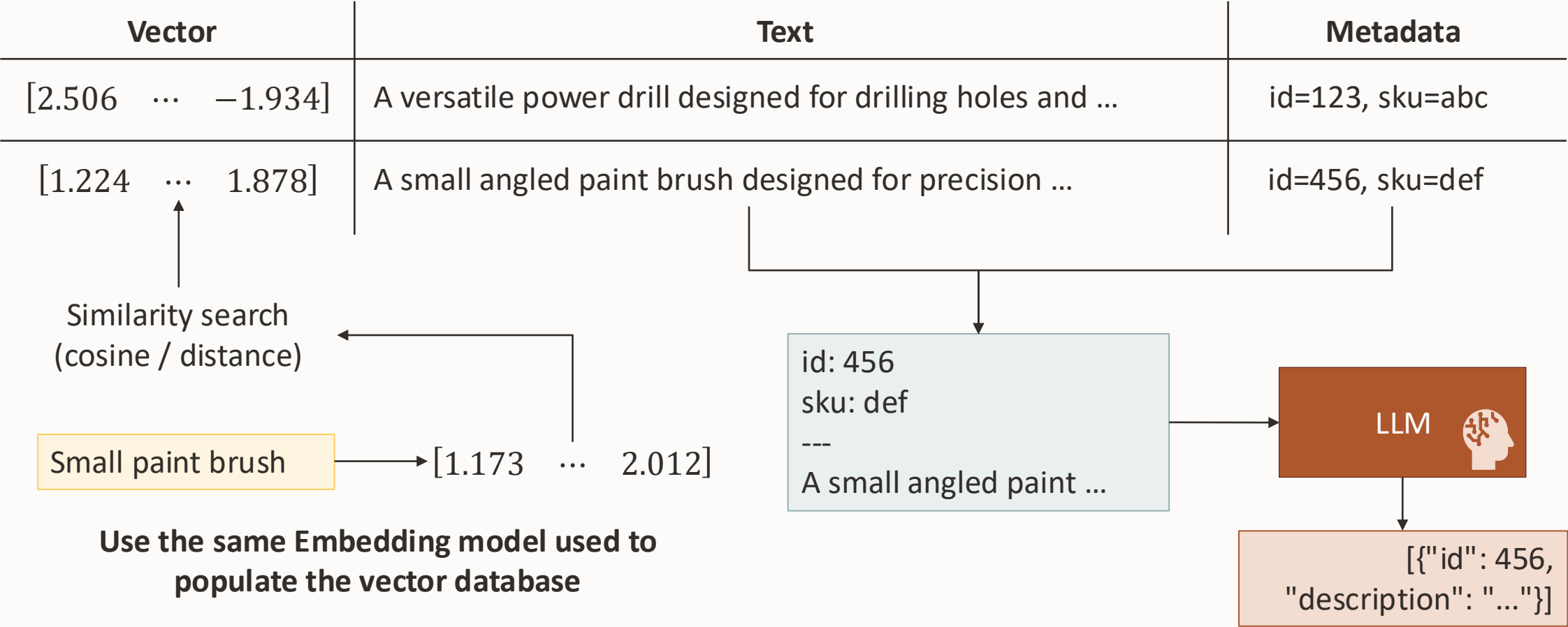
[2.506 ... -1.934]

Vector / Embedding

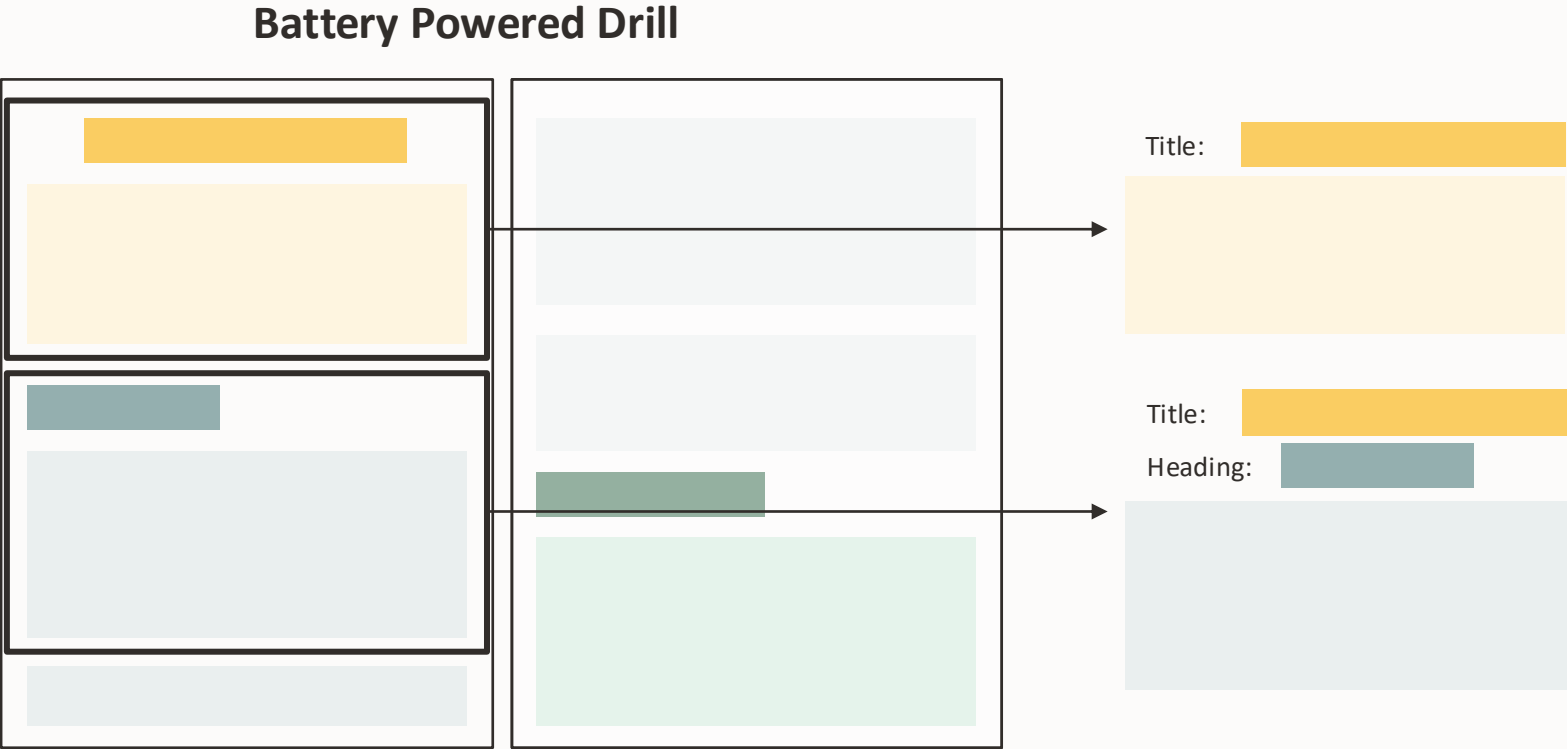
Similarity by embedding distance



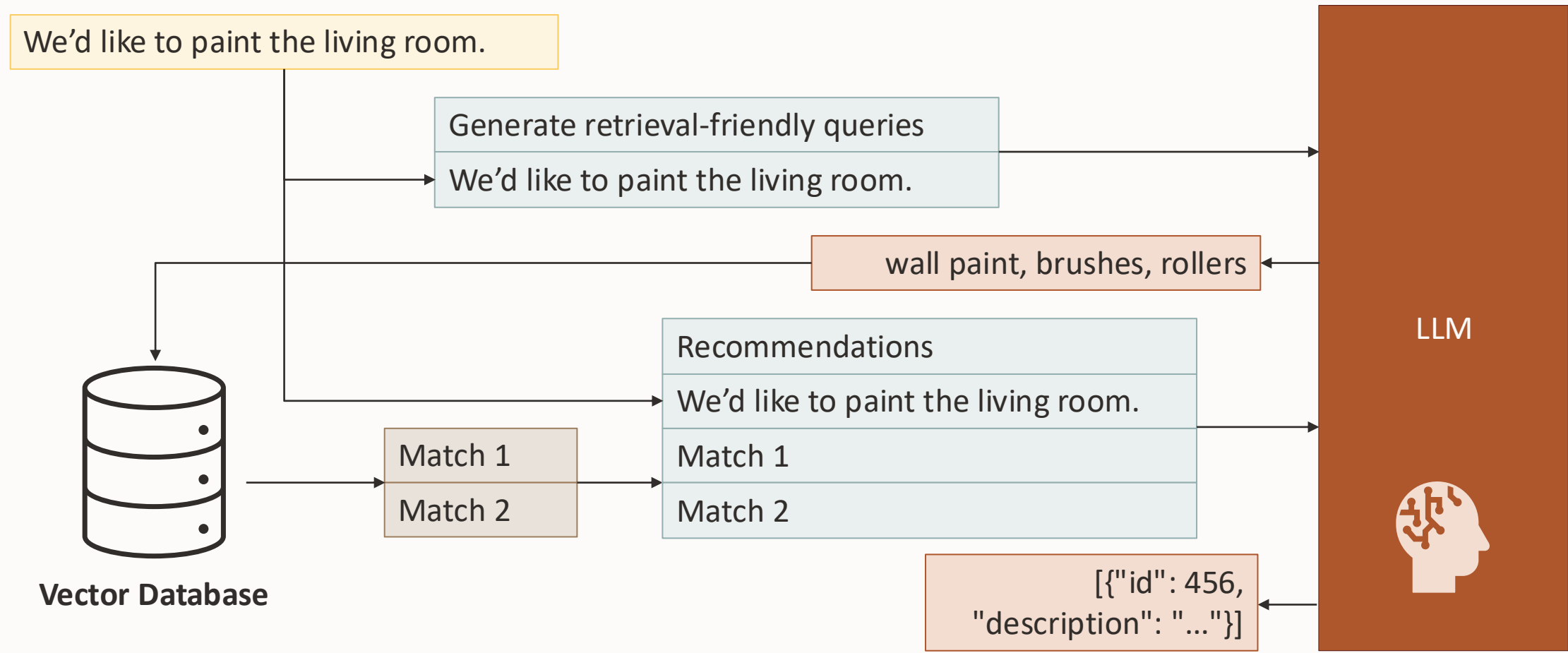
Storing and querying your knowledge



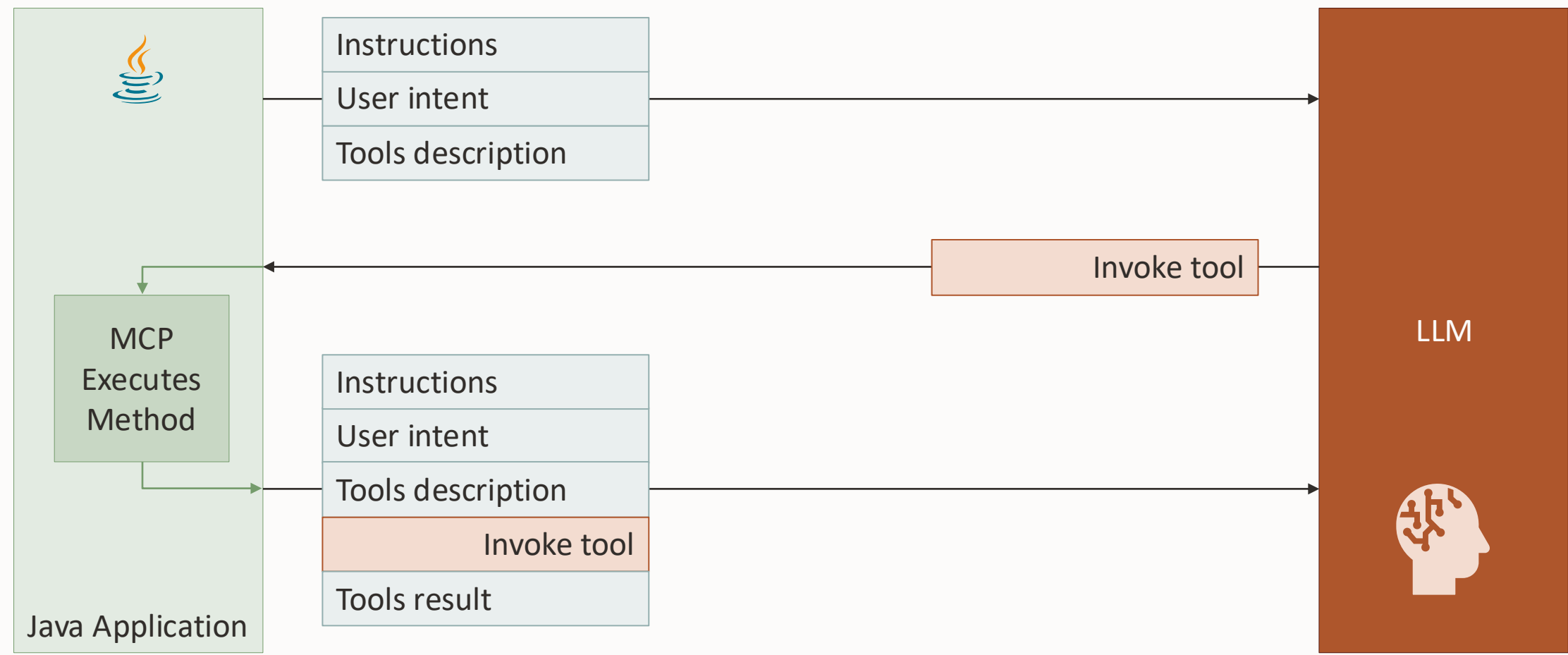
Preparing your data for RAG



From RAG to orchestration



Introducing Model Context Protocol (MCP)



Framework support for tool use

`@Component`

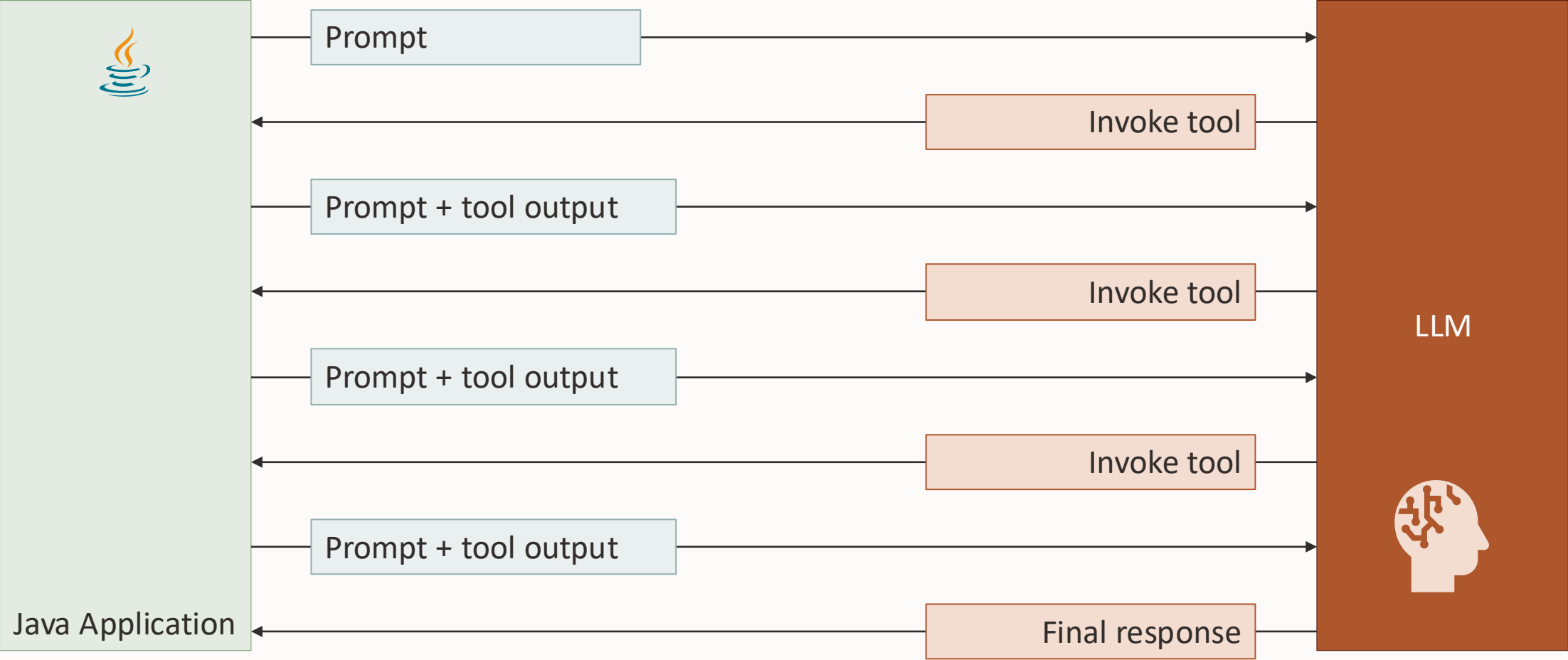
`class SearchTool {`

`@Tool(name = "search_product_catalogue",
 description = ""`

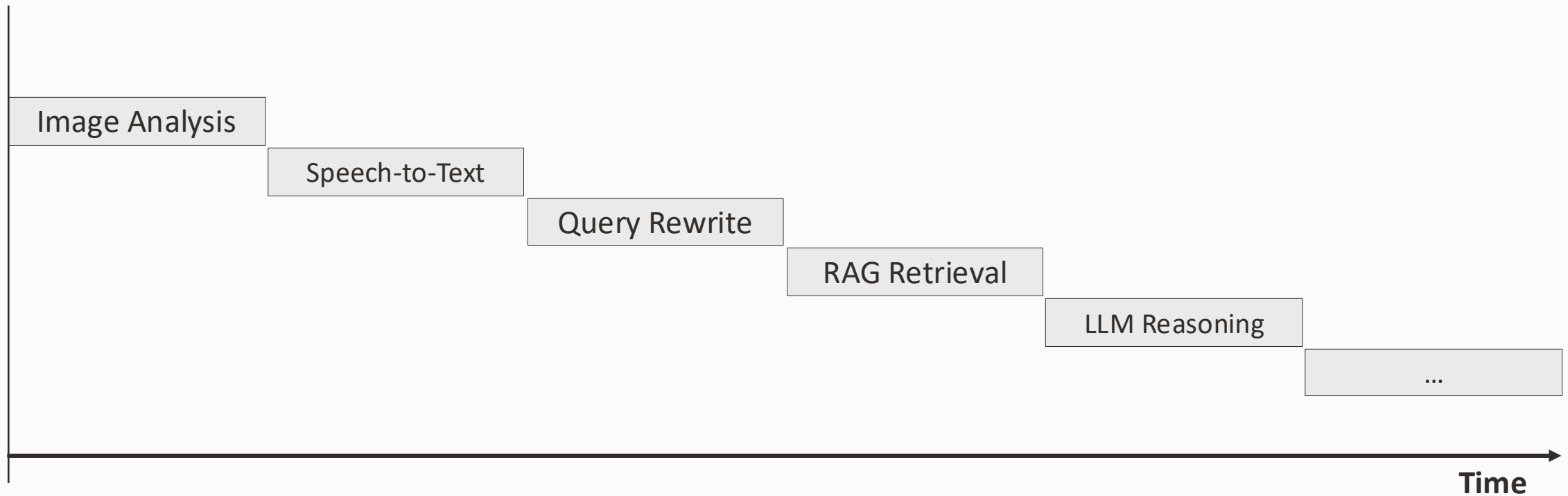
`Search the product catalogue and return matching \
 tools or products for the given list of names.""`

`List<Product> searchProductCatalogue(final @ToolParam(description = ""
 List of tool or product names to search for. \
 Each name may match multiple products."") List<String> names) {
 /* Closed for brevity */ } }`

Observability and cost matter more than ever

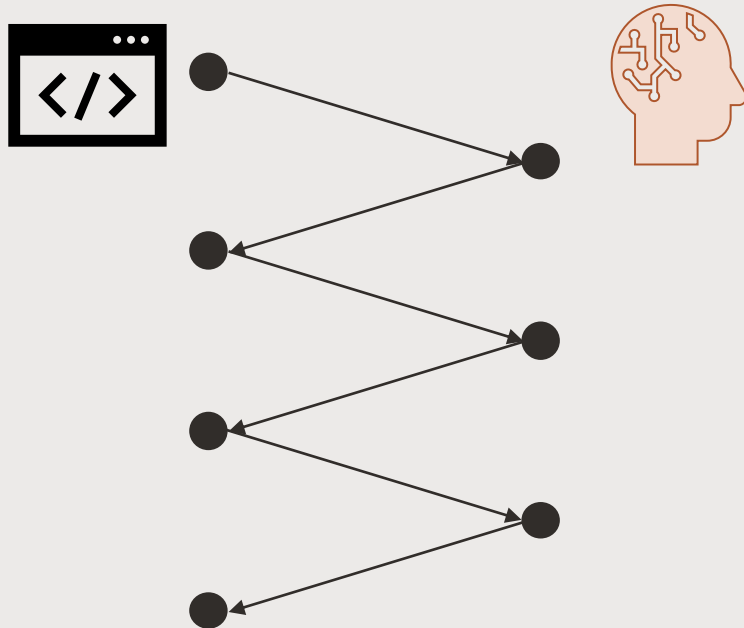


Latency is a real constraint

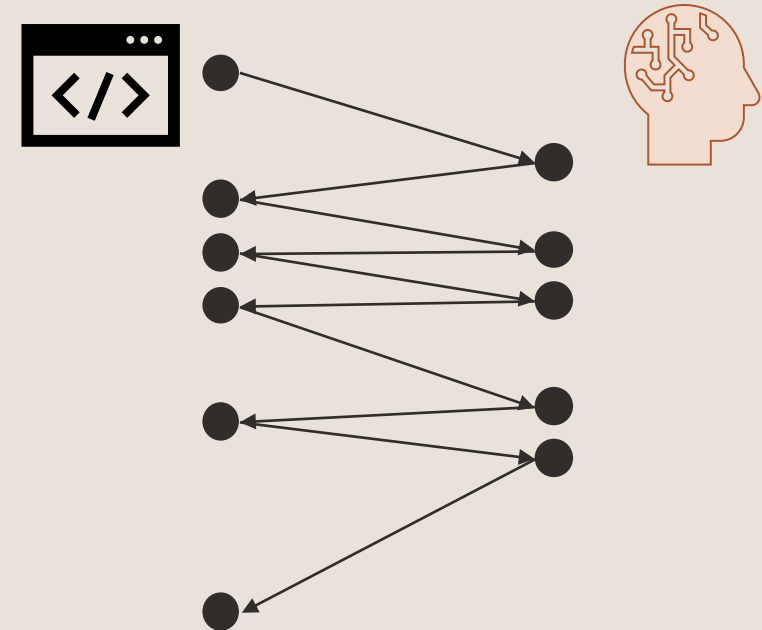


Trade-offs: RAG vs tool orchestration

RAG Consistent flow



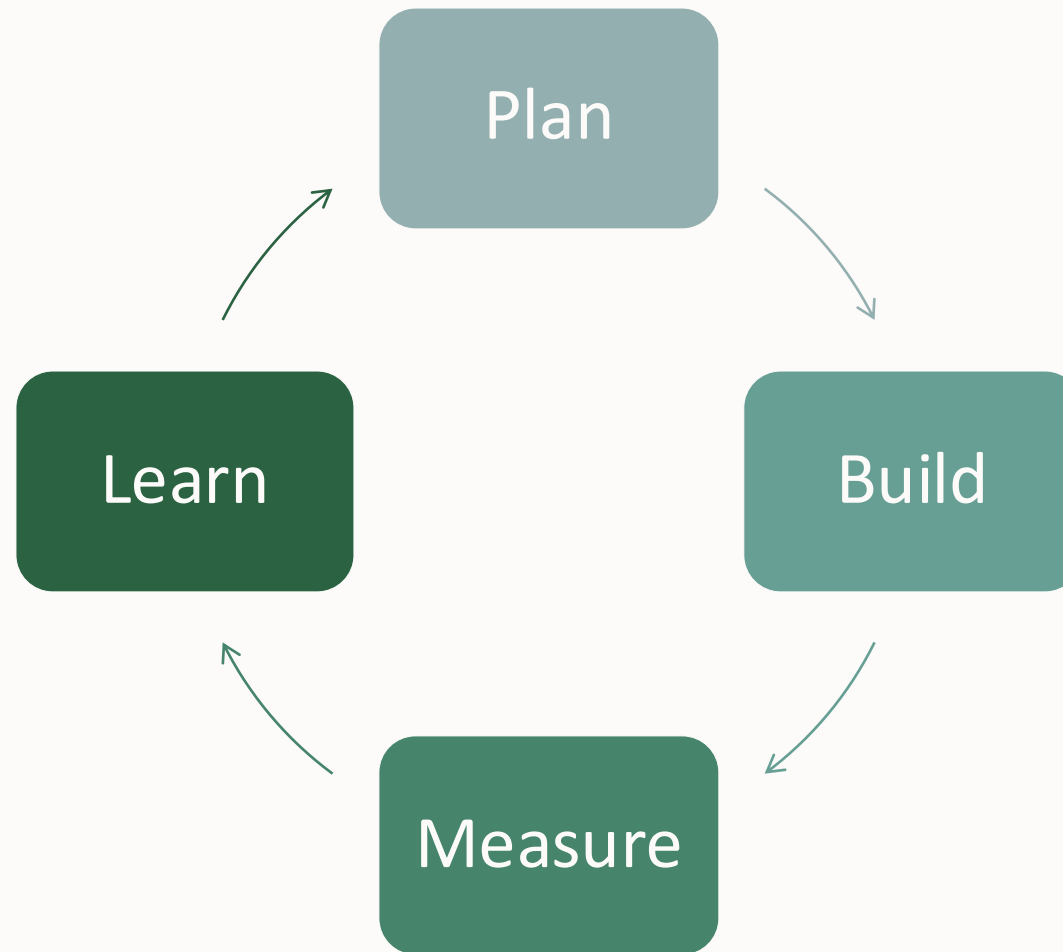
MCP Flow determined by model



What success actually looks like

$$\textit{Success} = \frac{\textit{Business Value}}{\textit{Costs}}$$

AI is just another product feature



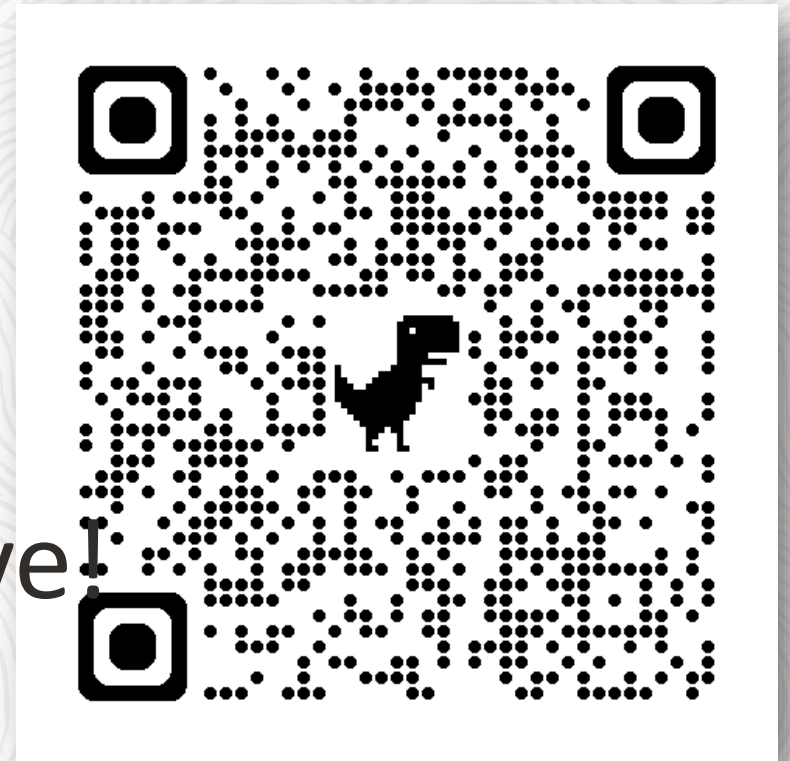
A final word of caution

Good engineering judgement still matters —
perhaps more than ever



Your feedback is invaluable to us!
Please share your thoughts
through our [brief survey](#) or
by email:
albert.attard@oracle.com.

Thank you for helping us improve!



Thank you

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